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ATTACHMENT ARRANGEMENT

Abstract

An attachment arrangement for attaching a structural element to a supporting element of a piece of furniture operable between multiple configurations, the attachment arrangement including a longitudinal guiding arrangement for attachment to one of the structural element or the supporting element, and an engaging arrangement for attachment to the other of the structural element or the supporting element. The engaging arrangement includes a first engagement member and a second engagement member operable to, when used in the piece of furniture, engage the longitudinal guiding arrangement to provide a translation and a rotation of the structural element in relation to the supporting element between a first configuration and a second configuration of the piece of furniture.

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Background/Summary

TECHNICAL FIELD

[0001] The present invention relates to an attachment arrangement and more precisely to an attachment arrangement for a piece of furniture allowing the piece of furniture to be operable in multiple configurations.

BACKGROUND

[0002] With the evolution of cities and the increase in prices of housing, many people are forced to live in a dwelling being smaller than they desire. There are also architectural and social movement (e.g. tiny/small-house movement) that advocates downsizing living spaces in order to simplify life and provide a more eco-friendly and less consumption oriented lifestyle.

[0003] With these smaller living accommodations, a need for smarter and more efficient usage of the living space becomes apparent. Living at a reduced area will make it challenging to furnish a home to have e.g. both a lounge area and a dining area. It may even be such that the only pieces of furniture available are a table and a bed where the bed has to function both as sofa, dining chair and sleeping arrangement. There are presently solutions available, e.g. day beds, murphy beds and futons, that may be configured to function as e.g. a sleeping arrangement and a sitting arrangement.

[0004] Multi-functional pieces of furniture are often bulky and the conversion from one configuration to another may be complicated and/or heavy. The mechanics involved in the conversion are generally complex and difficult to assemble making them costly to produce and transport.

SUMMARY

[0005] It is in view of the above considerations and others that the various embodiments of this disclosure have been made. The present disclosure therefor recognizes the fact that there is a need for improvement of the existing art described above.

[0006] An object of the present disclosure is to provide a new type of attachment arrangement which is improved over prior art and which eliminates or at least mitigates the drawbacks discussed above. More specifically, an object of the invention is to provide an attachment arrangement that is cost effective, intuitive and usable with pieces of furniture that are operable in multiple configurations. These objects are addressed by the technique set forth in the appended independent claims with preferred embodiments defined in the dependent claims related thereto.

[0007] In a first aspect, an attachment arrangement for attaching a structural element to a supporting element of a piece of furniture operable between multiple configurations is presented. The attachment arrangement comprises a longitudinal guiding arrangement for attachment to one of the structural element or the supporting element, and an engaging arrangement for attachment to the other of the structural element or the supporting element. The engaging arrangement comprising a first engagement means and a second engagement means operable to, when used in the piece of furniture, engage the longitudinal guiding arrangement to provide a translation and a rotation of the structural element in relation to the supporting element between a first configuration and a second configuration of the piece of furniture.

[0008] In one variant, when the attachment arrangement is used in the piece of furniture, the first engagement means is configured to provide a pivot point for rotation from the first configuration to the second configuration, and the second engagement means is configured to provide a pivot point for rotation from the second configuration to the first configuration. This is beneficial as the attachment arrangement will be more intuitive to use as the translation and the rotation will be in the same directions regardless of from which configuration the piece of furniture is changed.

[0009] In one variant, the first engagement means and/or the second engagement means is a rolling engagement means. Rolling engagement means provide a smooth and convenient translation and rotation as friction is reduced. Also, wear and tear of the engagement means and guiding arrangement will be reduced as friction is decreased.

[0010] In one variant, the first engagement means and/or the second engagement means is a sliding engagement means. A sliding engagement means is beneficial as it is comparably cheap and will reduce the cost of the attachment arrangement.

[0011] In one variant, the longitudinal guiding arrangement is provided with a first longitudinal member operable to, when used in the piece of furniture, support the engaging arrangement.

[0012] In one variant, the longitudinal guiding arrangement is further provided with a second longitudinal member arranged substantially parallel to the first longitudinal member. The longitudinal guiding arrangement is operable to, when used in the piece of furniture, sandwich one of the engagement means of the engaging arrangement between the first longitudinal member and the second longitudinal member during the translation of the structural element in relation to the supporting element. This is beneficial as it allows the engaging arrangement to be guided with more precision increasing the ease of use of the attachment arrangement and also provides a sturdier engagement between the guiding arrangement and the engaging arrangement. This increases the sense of quality of the attachment arrangement and also reduces play between the guiding arrangement and the engaging arrangement.

[0013] In one variant, the second longitudinal member is shorter than the first longitudinal member providing a longitudinal opening in the longitudinal guiding arrangement through which, when used in a piece of furniture (i.e. when in engagement), the first engagement means and the second engagement means are operable to pass. This allows for an intuitive process of converting the piece of furniture between the different configurations.

[0014] In one variant, the attachment arrangement comprises two longitudinal openings, wherein one longitudinal opening is provided at each longitudinal end of the longitudinal guiding arrangement and a longitudinal distance between the longitudinal openings is the same as a longitudinal distance between the first engagement means and the second engagement means. This allows for an intuitive process of converting the piece of furniture between the different configurations.

[0015] In one variant, the first longitudinal member and/or the second longitudinal member is formed with a U-shaped cross-section. This is beneficial as the engagement between the engaging arrangement and the guiding arrangement may be provided with reduced play increasing the sturdiness of the piece of furniture.

[0016] In one variant, the longitudinal guiding arrangement is provided with a longitudinal stop member configured to prevent the engagement means from, when engaging the longitudinal guiding arrangement, travel beyond the stop member.

[0017] In a second aspect, a piece of furniture operable between multiple configurations is presented. The piece of furniture comprising a structural element, a supporting element and an attachment arrangement according to the first aspect configured to attach one side of the structural element to the supporting element and to provide a translation and a rotation of the structural element in relation to the supporting element between a first configuration and a second configuration of the piece of furniture.

[0018] In one variant, the longitudinal guiding arrangement or the engagement arrangement attached to the structural element is arranged off-centered to a thickness extension of the structural element. This is beneficial as it will allow the structural element to be placed at different heights depending on the configuration. For instance, when a person is using the piece of furniture being in contact with the structural element, a distance from the attachment arrangement to the person will be different depending on the configuration of the piece of furniture.

[0019] In one variant, the piece of furniture comprises two attachment arrangements, one arranged

at each side of the structural element and configured to attach the structural element to the supporting element. This provides a stable and sturdy attachment of the structural element to the supporting element and reduce a risk of skewing of the structural element during reconfiguration of the piece of furniture.

[0020] In one variant, the structural element is formed with a chamfer at one end along the extension of the attachment arrangement. This is beneficial as parts of the structural element may be hidden and fitted behind parts of the supporting element in some configurations of the piece of furniture.

[0021] In one variant, the one or more longitudinal guiding means are attached to the supporting element.

[0022] In one variant, the piece of furniture is a seating furniture comprising one or more seats and one or more backrests.

[0023] In one variant, at least one of said one or more seats is a structural element attached to the supporting element by the attachment arrangement.

[0024] In one variant, at least one of said one or more backrests is a structural element attached to the supporting element by the attachment arrangement.

[0025] In one variant, one of the first configuration or the second configuration is a lounge configuration.

[0026] In one variant, one of the first configuration or the second configuration is a dining configuration.

[0027] In one variant, one of the first configuration or the second configuration is a sleeping configuration.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0028] Embodiments of the invention will be described in the following; references being made to the appended diagrammatical drawings which illustrate non-limiting examples of how the inventive concept can be reduced into practice.

[0029] FIGS. **1a-c** are perspective views of a piece of furniture operable between multiple configurations according to embodiments of the present disclosure;

[0030] FIGS. **2a-d** are schematic side views of an attachment arrangement according to embodiments of the present disclosure;

[0031] FIGS. **3a-d** are schematic side views of an attachment arrangement according to embodiments of the present disclosure;

[0032] FIG. **4** is a perspective view of an engaging arrangement according to embodiments of the present disclosure;

[0033] FIG. **5** is a perspective view of a guiding arrangement according to embodiments of the present disclosure;

[0034] FIG. **6** is a perspective view of a guiding arrangement in engagement with an associated engaging arrangement according to embodiments of the present disclosure;

[0035] FIGS. **7a-d** are partial perspective views of a piece of furniture comprising an attachment arrangement according to embodiments of the present disclosure; and

[0036] FIG. **8** is a partial perspective view of a piece of furniture comprising an attachment arrangement according to embodiments of the present disclosure.

DETAILED DESCRIPTION

[0037] Hereinafter, certain embodiments will be described more fully with reference to the accompanying drawings. The invention may, however, be embodied in many different forms and should not be construed as limited to the embodiments set forth herein; rather, these embodiments

are provided by way of example so that this disclosure will be thorough and complete, and will fully convey the scope of the invention, such as it is defined in the appended claims, to those skilled in the art.

[0038] The term “coupled” is defined as connected, although not necessarily directly, and not necessarily mechanically. Two or more items that are “coupled” may be integral with each other. The terms “a” and “an” are defined as one or more unless this disclosure explicitly requires otherwise. The terms “substantially,” “approximately,” and “about” are defined as largely, but not necessarily wholly what is specified, as understood by a person of ordinary skill in the art. The terms “comprise” (and any form of comprise, such as “comprises” and “comprising”), “have” (and any form of have, such as “has” and “having”), “include” (and any form of include, such as “includes” and “including”) and “contain” (and any form of contain, such as “contains” and “containing”) are open-ended linking verbs. As a result, a method that “comprises,” “has,” “includes” or “contains” one or more steps possesses those one or more steps, but is not limited to possessing only those one or more steps.

[0039] With reference to FIGS. **1a-c**, a piece of furniture **10** operable between multiple configurations according to embodiments of the present disclosure will be presented. The piece of furniture **10** of FIGS. **1a-c** is a sofa **10**, but the teachings of the present disclosure are applicable to many other types of furniture operable between multiple configurations such as chairs, beds etc. The piece of furniture **10** comprises a supporting element **30** configured to support a plurality of structural elements **20**. The supporting element **30** may be referred to as a frame **30** of the piece of furniture **10**. The supporting element **30** may be referred to as a load bearing element of the piece of furniture **10**. The supporting element **30** may be described as the element of the piece of furniture **10** that is fixed and not moved when changing the configuration of the piece of furniture **10**. The piece of furniture **10** further comprises a structural element **20** attached to the supporting element **30**. In the exemplary embodiment of FIG. **1a**, the two-seat sofa comprises a total of four structural elements **20**. Each seat is a structural element **20** and the backrest of each seat is a structural element **20**. Although not shown, also a headrest may be considered a structural element **20**. It should be emphasized that, for the teachings of the present disclosure, not all of the listed structural elements **20** of FIG. **1a** are needed to be present, for the teachings to be applicable, it is sufficient that one structural element **20** and one supporting element **30** is provided. In other words, in some embodiments, only the seats of a sofa are structural elements **20** according to the present disclosure. In some embodiments, not all of the seats of the piece of furniture **10** are structural elements **20** according to the present disclosure.

[0040] In FIG. **1a**, the seats of the piece of furniture **10** are illustrated in a first configuration (i). This first configuration (i) may, in some embodiments, be described as a lounge configuration. The lounge configuration is to mean that the piece of furniture **10** is configured, in the embodiment of the sofa, as a sofa for sitting and lounging. This generally implies that the seats are comparably low and typically incline downwards towards the backrest such that a person sitting on the sofa will be urged backwards into the sofa to a reclined position. In the lounge configuration, the seat may, in some embodiments, be softer compared to other configurations.

[0041] In FIG. **1b**, the same piece of furniture **10** as in FIG. **1a** is illustrated but with the left seat **20** and backrest **20** configured in a second configuration (ii). The second configuration (ii) may, in some embodiments, be described as a dining configuration. The dining configuration implies that the piece of furniture **10** is configured in the one of its configurations being most suitable for dining. Generally, when dining, a person prefers to sit more straight compared to when lounging. In FIG. **1b**, this is illustrated by a height of the left seat being increased and that an angle of the backrest of the sofa has been changed. The seat is, in the dining configuration, more horizontal than in the lounge configuration and the backrest is more vertical than in the lounge configuration. This allows a person sitting on the sofa to be seated more straight than in the lounge configuration and consequently be able to more conveniently reach e.g. a table during dining. Additionally, a seat for

dining is generally more firm than a seat for lounging which means that the dining configuration may be provided with a firmer seat than the lounge configuration.

[0042] It should be briefly mentioned that the angle of the backrest may be adjusted in accordance with the teachings of the present disclosure, but in some embodiments, the backrest may simply be pivotably mounted to the supporting element **30** at an end by the seat. The backrest may be pivoted about an axis parallel to a breadth extension of the sofa. In FIG. **1b**, the left seat is illustrated as pivoted forward and in order to keep it at this position, a neck rest (not shown) or similar may be folded between the backrest and the supporting element **30** to keep the backrest from pivoting backwards when a force is applied to the backrest, i.e. when a person sitting on the sofa is leaning backwards.

[0043] In FIG. **1c**, the same piece of furniture as in FIG. **1a** is illustrated but configured in an alternative second configuration (ii). This second configuration may, in some embodiments, be described as a sleeping configuration. As the name suggest, in the sleeping configuration, the piece of furniture **10** is configured as a bed. When sleeping, a substantially horizontal position is commonly preferred, and the seats are arranged, similarly to the dining position, substantially horizontally. One difference between the dining configuration and the sleeping configuration may be a firmness of the seat where a seat for dining is generally more firm than a seat or mattress for sleeping. In the dining configuration, it is generally preferred to maximize an area of the bed, and the backrests may be pushed away from the seats, into the sofa, at an end of the backrest by the seats.

[0044] In order to make the reconfiguration of the piece of furniture **10** as smooth and convenient as possible without having to use complex and expensive mechanical solutions, the inventors behind the present disclosure have realized that by using an attachment arrangement **100** as exemplified in FIG. **2a**, all these objectives are achieved. The attachment arrangement **100** comprises a longitudinal guiding arrangement **200** (guiding arrangement for short) and an engaging arrangement **300**. The longitudinal guiding arrangement **200** is configured to guide the engaging arrangement **300**. Preferably, the engaging arrangement **300** is attached to the structural element **20** of the piece of furniture **10** and the guiding arrangement **200** is attached to the supporting element **30**. However, as will be explained with reference to FIGS. **3a-d**, the opposite configuration is also possible and should be considered to form a part of the present disclosure, i.e. wherein the guiding arrangement **200** is attached to the structural element **20** of the piece of furniture **10** and the engaging arrangement **300** is attached to the supporting element **30**.

[0045] Generally, each of the longitudinal guiding arrangement **200** and the engaging arrangement **300** are fixedly attached to one of the structural element **20** and the supporting element **30**. By engagement of the engagement arrangement **300** with the longitudinal guiding arrangement **200**, the structural element **20** is movably attached to the supporting element **30** by the attachment arrangement **100**. The attachment between the structural element **20** and the supporting element **30** by the attachment arrangement **100** is such that the structural element **20** is supported by the supporting element **30**. The attachment arrangement **100** is configured such that a load of the structural element **20** is transferred to the supporting element **30**. To exemplify, if an attachment arrangement **100** according to the present disclosure is used to attach one side **20L**, **20R** (see FIG. **7c**) of a seat **20** of a sofa **10** to a frame **30** of the sofa **10**, that side of the seat would not require any additional load bearing features in order for the frame **30** to take the load of the seat **20**. As will further explained elsewhere in the present disclosure, it may be beneficial to provide one attachment arrangement **100** at each side (**20L**, **20R** of the seat **20** of the sofa **10**).

[0046] In FIG. **2a**, a cross-sectional view of the attachment arrangement **100** is shown where the engaging arrangement **300** is attached to the structural element **20** (not shown in FIG. **2a**) and the guiding arrangement **200** is attached to the supporting element **30** (not shown in FIG. **2a**) of the piece of furniture **10**. The engaging arrangement **300** comprises a first engagement means **310** and a second engagement means **320**. The engagement means **310**, **320** are spaced apart by a

longitudinal distance L.sub.300. The longitudinal distance L.sub.300 may be determined based on a configuration of the structural element **20** or the supporting element **30** that the engaging arrangement **300** is attached to. The engagement means **310**, **320** may be directly attached to the structural element **20** or to the supporting element **30**, or, as shown in FIG. **2a**, interconnected by a interconnecting means **330**. The engagement means **310**, **320** may be connected to the element **20**, **30** via the connection means **330**. The interconnecting means **330** may be any suitable interconnecting element such as a bar, strip or rod with any suitable profile and made from any suitable material such as plastic, wood or metal. The interconnecting means **330** is shown in FIG. **2a** as having a first longitudinal surface **331** facing vertically up, i.e. away from the guiding arrangement **200**. An opposite second longitudinal surface **332** of the interconnecting means **330** is facing vertically down in FIG. **2a**, i.e. towards the guiding arrangement **200**. In FIG. **2a**, the attachment arrangement **100** is shown in the first configuration (i), i.e. the piece of furniture **10** the attachment arrangement **100** is forming a part of (is comprised in) is configured in its first configuration (i). Assume for simplicity of explanation, that the structural element **20** to which the engaging arrangement **300** in FIGS. **2a-d** is attached, is a seat of a sofa and that the first configuration (i) is a lounge configuration. Assume further that the front of the sofa is, as shown in FIG. **2a**, towards the first engagement means **310** and the back of the sofa is towards the second engagement means **320**.

[0047] The transition of the attachment arrangement **100** (the piece of furniture **10**) to the second configuration (ii) is illustrated in FIGS. **2b** and **2c**. The transition may be described as a transformation, a reconfiguration, a change of mode, a change of operable configuration etc. In FIG. **2b**, the transition is initiated by a translation T and/or a rotation R of the engaging arrangement **300**, i.e. of the structural element **20** exemplified by the seat of a sofa. In this embodiment, the order of initiation of the translation T and rotation R is of no particular importance. Preferably, the seat is lifted at the front of the sofa, which means that the first engagement means **310** is lifted from the guiding arrangement **200** causing a rotation R about the second engagement means **320**. Before, after or in conjunction with this, the back of the seat is pulled towards the front of the sofa which means that the second engagement means **320** is longitudinally guided along the guiding arrangement **200** towards the front of the sofa, i.e. the translation T.

[0048] As seen in FIG. **2c**, the translation T is continued until the second engagement means **320** is at a longitudinal end of the longitudinal guiding arrangement **200**. Similarly, the rotation R continues until the first engagement means **310** is once more in contact with (or at least proximal to) the guiding arrangement **200**.

[0049] In FIG. **2d**, attachment arrangement **100** is shown when the transition from the first configuration (i) into the second configuration (ii) has been completed. Compared to first configuration (i) of FIG. **2a**, the second configuration (ii) is rotated 180 degrees about the engagement means **310**, **320**. As a result of the rotation R and the translation T, the first engagement means **310** and the second engagement means **320** have substantially changed positions with each other in relation to the guiding arrangement **200**. The rotation R was illustrated in FIGS. **2b** and **2c** as a rotation R about the second engagement means **310**, but the rotation R may very well be about the first engagement means **310**. In FIG. **2d**, the interconnecting means **330** is shown as having the second longitudinal surface **332** facing vertically up, i.e. away from the guiding arrangement **200**. The opposite first longitudinal surface **331** of the interconnecting means **330** is facing vertically down, i.e. towards the guiding arrangement **200**.

[0050] In FIGS. **3a-d**, cross-sectional views of an alternative arrangement of the attachment arrangement **100** is shown. In this arrangement, the engaging arrangement **300** is attached to the supporting element **20** (not shown in FIGS. **3a-d**) and the guiding arrangement **200** is attached to the structural element **20** (not shown in FIGS. **3a-d**) of the piece of furniture **10**. The transition from the first configuration (i), as shown in FIG. **3a**, to the second configuration (ii), as shown in

FIG. 3d, is analog to that described in connection with FIGS. 2a-d in that a translation T and a rotation R is executed on the supporting element 20. As the guiding arrangement 200 is attached to the supporting element 30 in this arrangement of the attachment arrangement 100, it is the guiding arrangement 200 that is subjected to the translation T and the rotation R. Assuming once more that the attachment arrangement 100 is comprised in a sofa and used to attach a structural element 20 in the form of a seat of the sofa to a supporting element 30 in the form a frame of the sofa. Assume further that the first engaging means 310 is arranged at the back of the sofa, i.e. towards the backrest of the sofa, and the second engaging means 320 is arranged at front of the sofa. In the embodiment of the attachment arrangement 100 shown in FIGS. 3a-d, the first engagement means 310 and the second engagement means 320 are attached directly to the supporting element 30 of the piece of furniture 10.

[0051] As seen in FIG. 3b, the transition is initiated by a translation T and/or a rotation R of the guiding arrangement 200, i.e. of the structural element 20 exemplified by the seat of a sofa. In this embodiment, as in the embodiment described above, the order of initiation of the translation T and rotation R is of no particular importance. Preferably, the seat is pulled forward out of the sofa causing the translation T of the guiding arrangement 200 such that it no longer is supported by the first engaging means 310. After this, the seat is lifted at the front of the sofa, which means that the guiding arrangement 200 is rotated R about the second engagement means 320, i.e. the rotation R.

[0052] As seen in FIG. 3c, the rotation R and the translation T is continued until the guiding arrangement 200 is arranged at a side of the second engagement means 320 facing away from the first engagement means 310.

[0053] In FIG. 3d, attachment arrangement 100 is shown after the transition from the first configuration (i) into the second configuration (ii) has been completed. Compared to first configuration (i) of FIG. 3a, in the second configuration (ii), the guiding arrangement 200 is rotated 180 degrees.

[0054] The arrangement illustrated in FIGS. 2a-d wherein the engaging arrangement 300 is attached to the structural element 20 is the preferred arrangement, the skilled person will appreciate that the opposite arrangement shown in FIGS. 3a-d wherein the guiding arrangement 200 is attached to the structural element 20 is also possible. The focus of the present disclosure will be on the first arrangement, but the skilled person will readily understand, after contemplating the teachings herein, which embodiments and features that are also usable with the latter arrangement. In addition to this, the skilled person will, after digesting the present disclosure, readily be able to make any suitable modifications to the latter arrangement in order to make it workable with specific features, even when those suitable modifications are not explicitly presented herein.

[0055] In the description given with reference to FIGS. 2a-d and 3a-d, the structural element 20 is illustrated as being rotated clockwise. The rotation R and translation T described may very well repeated once more in the clockwise direction, i.e. starting by lifting an end of the seat (the structural element 20) distal to a back of the sofa. Repeating the translation T and rotation R would mean that the structural element 20 is rotated, as viewed in Figs. FIGS. 2a-d and 3a-d, clockwise 360° returning it to the orientation illustrated in FIGS. 2a and 3a. The same end result is accomplished by reversing the rotation R and translation T, i.e. performing the rotation R counter clockwise by lifting an end of the seat (the structural element 20) proximal to the back of the sofa. The references to clockwise and counter clockwise are in relation rotations about pivot points provided by the engagement means 310, 320 as seen in the views in FIGS. 2a-d and 3a-d.

[0056] In one embodiment, the transition from the first configuration (i) to the second configuration (ii) is performed by translation T rotation R in a first direction and the transition from the second configuration (ii) to the first configuration (ii) is performed by translation T and rotation R in a second direction, opposite the first direction. This means that the same engagements means 310, 320 will be the pivot point for the rotation R regardless if the transition is from the first configuration (i) to the second configuration (ii) or from the second configuration (ii) to the first

configuration (ii).

[0057] In a preferred embodiment, the transition from the first configuration (i) to the second configuration (ii) is performed by translation T and rotation R in the first direction and the transition from the second configuration (ii) to the first configuration (ii) is also performed by translation T and rotation R in the first direction. This means that one engagements means **310, 320** will provide a pivot for the rotation R of the structural element **20** in the transition from the first configuration (i) to the second configuration (ii), and another engagement means **310, 320** will provide a pivot point for the rotation R of the structural element **20** from the second configuration (ii) to the first configuration (ii). This is beneficial as the structural element **20** will always be lifted at the same end providing a more user friendly and intuitive setup and also the wear and tear of the engagement means **310, 320** will be substantially equal extending the product life time of the attachment arrangement **100**.

[0058] As understood from the embodiments presented, the attachment arrangement **100** comprises at least two engagement means **310, 320**. Further engagement means **310, 320** may very well be provided to e.g. divide the load of the structural element **20** between more contact points between the engaging arrangement **300** and the guiding arrangement **200**. However, in a preferred embodiment, the attachment arrangement **100** comprises two engagement means **310, 320** and, as will be understood when all embodiments are presented, two engagement means **310, 320** are compatible with all embodiments herein.

[0059] The engagement means **310, 320** may, in some embodiments be substantially identical, and in other embodiments, different.

[0060] In some embodiments, one or both (or more) engagement means **310, 320** is/are a rolling engagement means. A rolling engagement means implies that a component of the engagement means is rotatable with respect to the structural element **20** or supporting element **30** to which it is attached. The rotation may be provided by friction between the rotating component of the engagement arrangement **300** and the longitudinal guiding arrangement **200** when the translation T of the structural element **20** occurs. The rotating component may be a wheel connected to the structural element **20** or the supporting element **30** by e.g. ball bearings etc. The rolling engagement means may be described as rolling in its engagement with the guiding arrangement **200**.

[0061] In some embodiments, one or both (or more) engagement means **310, 320** is/are a sliding engagement means. A sliding engagement means implies that a component of the engagement means is slidably engaging the guiding arrangement **200** when the attachment arrangement **100** is used in a piece of furniture **10**. The sliding between the engagement means and the guiding arrangement **200** is provided when the translation T of the structural element **20** occurs. The sliding engagement means may be described as sliding in its engagement with the guiding arrangement **200**.

[0062] In FIG. 4, one embodiment of an engaging arrangement **300** according to the teachings of the present disclosure is presented. The engaging arrangement **300** comprises a first engagement means **310** and a second engagement means **320** separated by a longitudinal distance L.sub.300. The longitudinal distance L.sub.300 may be defined as a center-to-center (cc) distance between the first engagement means **310** and a second engagement means **320**. In the embodiment of FIG. 4, the engagement means **310, 320** are arranged on an interconnecting means **330**, but as previously mentioned, this is purely optional and the engagement means **310, 320** may attached directly to the structural element **20** or to the supporting element **30**. The first engagement means **310** comprises an engagement portion **312** connected to the interconnecting means **330** by a cylindrical portion **314**. Analogously, the second engagement means **320** also comprises an engagement portion **322** connected to the interconnecting means **330** by a cylindrical portion **324**. The engagement portions **312, 322** are, preferably, the parts of the engagement arrangement **300** that are in contact with the guiding arrangement **200**. The engagement portions **312, 322** may be fixedly attached to the

cylindrical portions **314, 324** in embodiments where the engagement means **310, 320** is a sliding engagement means **310, 320** as described above. In embodiments where the engagement means **310, 320** is a rolling engagement means **310, 320**, the engagement means **310, 320** may be attached to the cylindrical portion **314, 324** by means of e.g. ball bearings etc. as described above. Alternatively, a rolling engagement means **310, 320** may be provided by forming a rolling attachment between the cylindrical portion **314, 324** and the interconnecting means **330**, or the structural element **20** or supporting element **30** if applicable. In the latter example, the engagement portions **312, 322** are preferably fixed to the cylindrical portions **314, 324**. The cylindrical portions **314, 324** are not necessarily cylindrical in shape, but may be any suitable shape or form. Generally, a diameter of the engagement portions **312, 322** is larger than a diameter of the cylindrical portions **314, 324**.

[0063] In FIG. 5, one embodiment of a longitudinal guiding arrangement **200** is illustrated. The guiding arrangement **200** comprises a first longitudinal member **210**. The engaging arrangement **300** will generally engage the first longitudinal member **210**, i.e. the engagement means **310, 320**, and specifically the engagement portion **312, 322** (if used) of the engaging arrangement **300** will typically contact the first longitudinal member **210**. The first longitudinal member **210** may, in some embodiments, be provided with a guide rim **215** extending along the first longitudinal member **210** and being arranged at an angle to the first longitudinal member **210**. The guide rim **215** is configured to prevent the engagement means **310, 320** from exiting the first longitudinal member in one direction along the extension of the cylindrical portion **314, 324**. The structural element **20** or supporting element **30** that the guiding arrangement is attached to will prevent the engagement means **310, 320** from exiting the first longitudinal member in the other direction along the extension of the cylindrical portion **314, 324**. In embodiment with the guide rim **215** and when the engaging arrangement **300** is engaging the guiding arrangement **200**, the engagement portions **312, 322** of the engaging arrangements are sandwiched between the guide rim **215** and the one of the structural element **20** or supporting element **30** that the guiding arrangement **200** is attached to.

[0064] The guiding arrangement **200** may optionally be provided with longitudinal stop members **250, 260** (stop members for short) provided at longitudinal ends of the guiding arrangement **200**. A first stop member **250** may be provided at one longitudinal end of the guiding arrangement **200** and a second stop member **260** may be provided at the other longitudinal end of the guiding arrangement **200**. The stop members **250, 260** are configured to prevent the engagement means **310, 320** from traveling, during translation T, beyond the longitudinal extension of the first longitudinal member **210**. The stop members **250, 260** are preferably arranged substantially transverse (substantially perpendicular) to the first longitudinal member **210** towards the engaging arrangement (when in engagement). The stop members **250, 260** are beneficial as they prevent the translation T to go beyond the longitudinal extension of the guiding arrangement **200**. To exemplify, looking back to FIG. 2a and assuming that there is a first stop members **250** at the front of the sofa, i.e. to the left in FIG. 2a. This would mean that the only option to start a transition to the second configuration (ii) would be to lift the first engagement means **310** being adjacent to the first stop member **250** (not shown in FIG. 2a). As the translation T and rotation R continues, see FIG. 2c, the first stop member **250** (not shown in FIG. 2c) would prevent the second engagement means **320** to travel beyond the longitudinal extension of the guiding arrangement **200** such that the rotation R may be completed without risking a too long translation T. Returning to FIG. 5, the second stop member **260** may similarly be provided to prevent the engaging arrangement **300** to travel beyond the guiding arrangement **300** at the other longitudinal end of the guiding arrangement **300**.

[0065] The guiding arrangement **200** in FIG. 5 is further provided with an optional second longitudinal member **220**. The second longitudinal member **220** is substantially parallel to the first longitudinal member **210**. A distance between the first longitudinal member **210** and the second longitudinal member **220** is adapted such that the engagement means **310, 320** of the engaging

arrangement **300** may be sandwiched between the first longitudinal member **210** and the second longitudinal member **220** during the translation T. That is to say, the engagement means **310, 320** engaging the guiding arrangement **200** during the translation T is sandwiched between the first longitudinal member **210** and the second longitudinal member **220** for at least a portion of the translation T. In embodiments of the engaging arrangement **300** wherein the engagement means **310, 320** are provided with engagement portions **312, 322**, the distance between the first longitudinal member **210** and the second longitudinal member **220** is preferably slightly larger than the diameter of the engagement portions **312, 322** to allow translation T without undue friction and also permit a rolling engagement means to freely rotate between the longitudinal members **210, 220**. The second longitudinal member **220** may be arranged to prevent undesired rotation R of the engaging arrangement **300** when it is engaging the guiding arrangement **200**. Particularly, during rotation R, the second longitudinal member **220** prevents the engagement means **310, 320** currently out of engagement with the guiding arrangement **300** from being lifted from the guiding arrangement. In the case of the sofa in FIG. 2c, this means that the second engagement means **320** is retained at a substantially constant horizontal position throughout the transformation from the first configuration (i) to the second configuration (ii) as illustrated in FIGS. 2b and 2c.

[0066] In the embodiment in FIG. 5, the second longitudinal member **220** is longitudinally shorter than the first longitudinal member **210**. The second longitudinal member **220** is parallel to the first longitudinal member and arranged with a longitudinal center substantially aligned with a longitudinal center of the first longitudinal member **210**. This arrangement provides longitudinal openings **230, 240** in the guiding arrangement **200**. In embodiments comprising the longitudinal stop members **250, 260**, as the one in FIG. 5, the longitudinal openings **230, 240** are provided, between the longitudinal stop members **250, 260** and the second longitudinal member **220**. A longitudinal distance $L_{sub.200}$ between a first longitudinal opening **230** at one longitudinal end of the second longitudinal member **220** and a second longitudinal opening **240** at the other longitudinal end of the second longitudinal member **220** is preferably adapted to match the longitudinal distance $L_{sub.300}$ between the first engagement means **310** and the second engagement means **320**. Preferably, the longitudinal extension of the second longitudinal member **220** is shorter than a shortest longitudinal distance between the engagement means **310, 320**, i.e. the cc longitudinal distance L_{300} between the engagement means **310, 320** minus two times a radius of the engagement portions **312, 322**. This will permit the engaging arrangement **300** to be inserted into the guiding arrangement **200**, i.e. arranged in engagement with the first longitudinal member **210** of the guiding arrangement **200**, in a direction transversal to the longitudinal extension of the first longitudinal member **210** and the second longitudinal member **220**.

[0067] It should further be mentioned that the first longitudinal member **210** and/or the second longitudinal member **220** may be formed with a U-shaped cross-section providing guide rims **215** (only shown for the first longitudinal member **210**). Preferably, the U-shaped cross-section is adapted to engage the engagement portions **312, 322** of the engagement means **310, 320** effectively retaining the engagement means **310, 320** and preventing movement in a direction along the extension of cylindrical portions **314, 324** of the engagement means **310, 320**.

[0068] It should be mentioned that the features listed above with reference to guiding arrangement **200** illustrated in FIG. 5 may very well separated as individual features and must not be combined in one embodiment. That is to say, the stop members **250, 260** may be selected for an embodiment without having to include e.g. the second longitudinal member **220** and the second longitudinal member **220** may be selected for another embodiment without having to include the stop members **250, 260**, the guide rim **215** and/or the longitudinal openings **230, 240**. However, the embodiment of the guiding arrangement **200** illustrated in FIG. 5 provides excellent guiding for an associated engaging arrangement **300**.

[0069] In FIG. 6 the same corresponding guiding arrangement **200** as shown in FIG. 5 is depicted but in engagement with an associated engaging arrangement **300**. To exemplify the benefits of the

features of the attachment arrangement **100**, In FIG. **6**, the first engagement means **310** will be allowed to exit the guiding arrangement **200** through the first opening **230** whilst the second engagement means **320** will be kept sandwiched in the guiding arrangement. The second engagement means **320** will be permitted to, rolling or slidingly, move along the guiding arrangement **200** sandwiched between the first longitudinal member **210** and the second longitudinal member **220** during the translation T when changing configuration of to the piece of furniture **10**. The translation T will be stopped when the second engagement means **320** reaches the first longitudinal stop member **250** at which point a full 180° rotation R will allow the first engagement means **310** to once more come in contact with the first longitudinal member **210** through the second opening **240**.

[0070] The guiding arrangement **200** of the present disclosure may be formed from any suitable material. However, preferably the guiding arrangement **200** is made from metal. It may be beneficial to form the guiding arrangement from a sheet metal shaped (e.g. cut) such that it may be folded to form e.g. the first longitudinal member **210** and (if present) the second longitudinal member **220**, (if present) the guide rims **215** and/or (if present) the stop members **250**, **260**. This is beneficial as a strong, yet cost effective, guiding arrangement is provided.

[0071] With reference to FIGS. **7a-d**, the transformation, i.e. reconfiguration, change of mode, change of operable configuration etc., of a piece of furniture **10** in the form of a sofa from the first configuration (i) to the second configuration (ii) will be illustrated. The structural element **20** is illustrated as a seat, but may, as previously explained, be the backrest. The engaging arrangement **300** is attached to the structural element **20** and the guiding arrangement **200** is attached to the supporting element **30**. The engaging arrangement **300** is not attached centrally with regards to thickness extension h.sub.20, or height, of the structural element **20**. The engaging arrangement **300** is, in other words, arranged off-centered to the thickness extension h.sub.20 of the structural element **20**. This will result in different heights of the structural element **20** depending on the configuration of the piece of furniture **10**. That is to say, in the first configuration (i), a top surface of the seat may be at a first distance from a floor on which the sofa stands, and in the second configuration (ii), the seat may be at a second distance from the floor, wherein the second distance is longer than the first distance. A seated height of a person sitting in the sofa will be higher in the second configuration (ii) than in the first configuration (ii).

[0072] In FIG. **7b**, the transformation of the piece of furniture **10** is initiated by rotation R about the second engagement means **320**. This is accomplished by lifting the end of the structural element **20**, the seat, being closest to the first engagement means **310** such that the first engagement means **310** may pass through the first longitudinal opening **230**.

[0073] In FIG. **7c**, the rotation R about the second engagement means **320** is continued and the second engagement means **320** is translated T towards the first longitudinal opening **230**, i.e. towards the end of the guiding arrangement **200** at which the first engaging means **310** was located before the transformation from the first configuration (i) to the second configuration (ii) was initiated.

[0074] In FIG. **7d**, the piece of furniture is shown in the second configuration, i.e. with transformation from the first configuration (i) to the second configuration (ii) completed. The first engaging means **310** has entered the guiding arrangement **200** through the second longitudinal opening **240**.

[0075] As seen in FIGS. **7a-d**, the seat is provided with a chamfer **25** at one end of the seat, i.e. the structural element **20**. This chamfer **25** is beneficial when the attachment arrangement **100** is used in a seating furniture with a backrest. A seating furniture may describe any piece of furniture one may sit on such as, but not limited to, sofas, armchairs, loveseats, couches, chairs, stools etc. The chamfer **25** is especially beneficial in embodiments wherein the part of the attachment arrangement attached to the structural element **20** in the form of a seat is offset with regards to the thickness extension h.sub.20 of the structural element **20**. The chamfer **25** will fit under the backrest when

the seat is in a higher, i.e. raised sitting position. When the seat is in the lower sitting position, the chamfer **25** will allow the seat to extend into the sofa whilst still supporting the legs of a person sitting on the sofa.

[0076] With reference to FIG. **8**, a preferred embodiment of a piece of furniture **10** according to the teachings of the present disclosure will be presented. The view of FIG. **8** is a partial view of a two-seat sofa showing the sitting section of the sofa with padding removed. Each of the two seats of the sofa are operable in two different configurations, i.e. the seats are the structural elements **20**, as described herein. Each of the seats is attached to the frame of the sofa, i.e. the supporting element **30** as taught herein, by two attachment arrangements **100**, one at each side **20L**, **20R** of the seat. In FIG. **8**, the attachment arrangements **100** are shown as substantially equal (with the exception of the middle guiding arrangement **200** which is discussed in the following section) at both sides **20L**, **20R** of the structural element **20**. However, this is but one embodiment, and in some embodiments, the attachment arrangements **200** at each side **20L**, **20R** of the structural element **20** may be different. In one embodiment, only one of the attachment arrangements **100** of a structural element **20** may be provided with the second longitudinal member **220**, the stop members **250**, **260** and/or the longitudinal openings **230**, **240**.

[0077] As illustrated in FIG. **8**, the guiding arrangement **200** arranged between the seats of the sofa may be a doubled guiding arrangement **200**. The doubled guiding arrangement **200** is illustrated in the magnified section in FIG. **8**. The doubled guiding arrangement **200** may be formed by any guiding arrangement **200** presented herein, and the one shown in FIG. **8** is a doubled guiding arrangement **200** based on the guiding arrangement shown in FIG. **5**. Basically, the doubled guiding arrangement **200** is a combination of two guiding arrangements, mutually mirrored (or rotated if they are symmetrical) in a plane extending between the first longitudinal member **210** and the second longitudinal member **220**. The doubled guiding arrangement **200** is beneficial as it allows two engaging arrangement **300** to engage it simultaneously, one at each side.

[0078] Modifications and other variants of the described embodiments will come to mind to one skilled in the art having benefit of the teachings presented in the foregoing description and associated drawings. Therefore, it is to be understood that the embodiments are not limited to the specific example embodiments described in this disclosure and that modifications and other variants are intended to be included within the scope of this disclosure. For example, while embodiments of the invention have been described with reference mainly to a seat of a sofa with the guiding arrangement attached to the frame of the sofa, persons skilled in the art will appreciate that the embodiments of the invention can equivalently be applied to other parts of furniture and with the position of the guiding arrangement and the engaging arrangement reversed. Furthermore, although specific terms may be employed herein, they are used in a generic and descriptive sense only and not for purposes of limitation. Therefore, a person skilled in the art would recognize numerous variations to the described embodiments that would still fall within the scope of the appended claims. Furthermore, although individual features may be included in different claims (or embodiments), these may possibly advantageously be combined, and the inclusion of different claims (or embodiments) does not imply that a combination of features is not feasible and/or advantageous. In addition, singular references do not exclude a plurality. Finally, reference signs in the claims are provided merely as a clarifying example and should not be construed as limiting the scope of the claims in any way.

Claims

1. An attachment arrangement for attaching a structural element to a supporting element of a piece of furniture operable between multiple configurations, the attachment arrangement comprising: a longitudinal guiding arrangement for attachment to one of the structural element or the supporting element, and an engaging arrangement for attachment to the other of the structural element or the

- supporting element; the engaging arrangement comprising a first engagement member and a second engagement member operable to, when used in the piece of furniture, engage the longitudinal guiding arrangement to provide a translation and a rotation of the structural element in relation to the supporting element between a first configuration and a second configuration of the piece of furniture.
2. The attachment arrangement of claim 1, when used in the piece of furniture, the first engagement member is configured to provide a pivot point for rotation from the first configuration to the second configuration, and the second engagement member is configured to provide a pivot point for rotation from the second configuration to the first configuration.
 3. The attachment arrangement of claim 1, wherein the first engagement member and/or the second engagement member is a rolling engagement member.
 4. The attachment arrangement of claim 1, wherein the first engagement member and/or the second engagement member is a sliding engagement member.
 5. The attachment arrangement of claim 1, wherein the longitudinal guiding arrangement is provided with a first longitudinal member operable to, when used in the piece of furniture, support the engaging arrangement.
 6. The attachment arrangement of claim 5, wherein the longitudinal guiding arrangement is further provided with a second longitudinal member arranged substantially parallel to the first longitudinal member, wherein the longitudinal guiding arrangement is operable to, when used in the piece of furniture, sandwich one of the engagement member of the engaging arrangement between the first longitudinal member and the second longitudinal member during the translation of the structural element in relation to the supporting element.
 7. The attachment arrangement of claim 6, wherein the second longitudinal member is shorter than the first longitudinal member providing a longitudinal opening in the longitudinal guiding arrangement through which, when used, the first engagement member and the second engagement member are operable to pass.
 8. The attachment arrangement of claim 7, comprising two longitudinal openings, wherein one longitudinal opening is provided at each longitudinal end of the longitudinal guiding arrangement and a longitudinal distance between the longitudinal openings is the same as a longitudinal distance between the first engagement member and the second engagement member.
 9. The attachment arrangement of claim 6, wherein the first longitudinal member and/or the second longitudinal member is formed with a U-shaped cross-section.
 10. The attachment arrangement of claim 1, wherein the longitudinal guiding arrangement is provided with a longitudinal stop member configured to prevent the engagement member from, when engaging the longitudinal guiding arrangement travel beyond the stop member.
 11. A piece of furniture operable between multiple configurations, the piece of furniture comprising the structural element, the supporting element and the attachment arrangement according to claim 1 configured to attach one side of the structural element to the supporting element and to provide the translation and the rotation of the structural element in relation to the supporting element between the first configuration and the second configuration of the piece of furniture.
 12. The piece of furniture of claim 11, wherein the longitudinal guiding arrangement or the engagement arrangement attached to the structural element is arranged off-centered to a thickness extension of the structural element.
 13. The piece of furniture of claim 11, comprising two attachment arrangements, one arranged at each side of the structural element and configured to attach the structural element to the supporting element.
 14. The piece of furniture of claim 11, wherein the structural element is formed with a chamfer at one end along an extension of the attachment arrangement.
 15. The piece of furniture according to claim 11, wherein one or more of the longitudinal guiding arrangement are attached to the supporting element.

16. The piece of furniture according to claim 11, wherein the piece of furniture is a seating furniture comprising one or more seats and one or more backrests.

17. The piece of furniture according to claim 16, wherein at least one of said one or more seats is the structural element attached to the supporting element by the attachment arrangement.

18. The piece of furniture according to claim 16, wherein at least one of said one or more backrests is the structural element attached to the supporting element by the attachment arrangement.
